19 — Two Aux Engines: the only scenario that exercises Level 2

Proves · The only scenario where Level 2 does visible work — and Level 3 with it.

Mechanics this scenario turns on

- Level 2 redistributes the hotel load between the shaft generator and the auxiliaries, looking for a cheaper split. It can only find one when there is something to redistribute — with the SG at its ceiling and a single aux, it returns zero.
- Level 3 (DRC) damps the hotel/mission swing: `variation × 0.8`, minus whatever the battery already shaved, so the same kilowatt is never counted twice.
- Level 2 and Level 3 are computed for **Transit only** (decision D4/Q5 — no workbook counterpart elsewhere). Other modes get an empty result, and the pinned-baseline radio does not reach them.
- The shaft generator is filled before any auxiliary starts (the main engine is already turning), and its output is a load **on** that main engine — which is why the ME figure exceeds propulsion. SG capacity scales with the number of running MEs.
- Baseline rule: **no battery → the worst combination** (`count - 1`); **battery active → the third from worst** (`Math.Max(0, count - 3)`). It models what the ship is assumed to do today.

Panels described below · Why Level 2 was invisible · The plant, built for exactly that · Level 1 · Level 2 — the point of the scenario · Result · Open question for the workbook

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 11.

Why it exists. Level 2 produced **zero savings in all 18 original scenarios**. A whole optimization level — the recursive sweep — had no end-to-end coverage: break it so it returns zero and every snapshot would still pass.

Why Level 2 was invisible

Read the aux SFOC curve (engine 8) and it falls monotonically:

```
10 % → 228.47      50 % → 198.07      74–80 % → 193.00      100 % → 194.63
```

More load is always better. Level 1 already evaluates every engine **count** and picks the cheapest, so "run fewer engines harder" is a decision Level 1 has made before Level 2 is asked.

Level 2's only remaining lever is an **asymmetric** split — and that needs a demand a single engine cannot legally carry.

The plant, built for exactly that

```
ME 2 × 8 500 · no SG · AE 2 × 2 000 · propulsion 8 000 · hotel 2 000 · 5 000 h
```

- No shaft generator ⇒ the aux side must carry the whole 2 000 kW hotel load.
- 1 aux engine ⇒ 2 000 / 2 000 = **100 %** — above the 90 % ceiling, rejected.
- 2 aux engines ⇒ 50 % each. Legal, and squarely in the steep part of the curve.

Level 1

Two valid combinations (ME = 0 is invalid in Transit, aux count is forced to 2):

#	ME	SG	AE	FOC t/h
0	2	off	2	1.734727 ← optimal
1	1	off	2	1.772322 ← baseline

Baseline **8 861.6 t/yr** · L1 savings **187.97 t/yr**.

Level 2 — the point of the scenario

Level 1 handed the aux side an even split: 2 × 1 000 kW = 50 % each. The sweep searches 10 %…90 % in 2 % steps and finds:

```
AE₁   200 kW  @ 10 %   sfoc 228.47
AE₂ 1 800 kW  @ 90 %   sfoc 193.78
```

Hand-check

```
even split : 2 000 kW × 198.07 g/kWh                 = 0.396140 t/h
asymmetric :   200 × 228.47  +  1 800 × 193.78       = 0.394498 t/h
difference × 5 000 h                                 ≈ 8.21 t/yr
reported                                             = 8.14 t/yr
```

Agreement to within rounding of the curve read-off. **The exact figure is pending reference verification** — it came from the code, not the workbook.

Result

L1	187.97
L2	**8.14**
L3	45.25
Advanced / Pro / Premium	187.97 / **196.11** / **241.36**

The first scenario in the suite where all three tiers differ.

Open question for the workbook

The model's answer is one generator idling at 10 % and one at 90 %. A crew would more likely run a **single** generator at 100 % — which the 90 % ceiling forbids. If the workbook disagrees with 10/90, the floor or the ceiling is mis-set. Worth confirming before this figure is treated as correct.

Takeaway: Level 2 only has room when a single engine cannot legally carry the aux demand. That is a narrow window, and no scenario had ever landed in it.